SHETH MVLU COLLEGE
SUBJECT:- AI

Practical 8

Aim:- K-Nearest Neighbors (K-NN)

- Implement the K-NN algorithm for classification or regression.
- Apply the K-NN algorithm to a given dataset and predict the class or value for test data.
- Evaluate the accuracy or error of the predictions and analyze the results.

INPUT:-

Ai KNN.py - D:/Anees S110/Anees T110/Ai KNN.py (3.13.2)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, ConfusionMatrixDisplay

# Load Dataset
print("Loading dataset...")
df = pd.read_csv('Airline Dataset.csv')
print("Dataset loaded successfully! Total rows:", len(df))

# Data Preprocessing & Encoding
df_clean = df[['Age', 'Gender', 'Continents', 'Flight Status']].dropna().copy()

le_gender = LabelEncoder()
le_continents = LabelEncoder()
le_target = LabelEncoder()

df_clean['Gender'] = le_gender.fit_transform(df_clean['Gender'])
df_clean['Continents'] = le_continents.fit_transform(df_clean['Continents'])
df_clean['Flight Status'] = le_target.fit_transform(df_clean['Flight Status'])

X = df_clean[['Age', 'Gender', 'Continents']]
y = df_clean['Flight Status']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
```

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```python
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Model Implementation & Tuning
print("\nTraining K-NN Model...")
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
print("Model trained successfully!")

# Model Evaluation & Prediction
y_pred = knn.predict(X_test_scaled)

print("\n EVALUATION RESULTS")
print("Overall Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=le_target.classes_))

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

print("\n PREDICTION ON TEST DATA")

# FIX: Passed DataFrame with feature names instead of a plain list
sample_df = pd.DataFrame(
    [[45, le_gender.transform(['Female'])[0], le_continents.transform(['Asia'])[0]]],
    columns=['Age', 'Gender', 'Continents']
)
```

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Ai KNN.py - D:/Anees S110/Anees T110/Ai KNN.py (3.13.2)

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```python
print("\n PREDICTION ON TEST DATA")

# FIX: Passed DataFrame with feature names instead of a plain list
sample_df = pd.DataFrame(
    [[45, le_gender.transform(['Female'])[0], le_continents.transform(['Asia'])[0]]],
    columns=['Age', 'Gender', 'Continents']
)

sample_scaled = scaler.transform(sample_df)
pred = knn.predict(sample_scaled)
print("Input Sample: Age=45, Gender=Female, Continent=Asia")
print("Predicted Flight Status:", le_target.inverse_transform(pred)[0])

print("\n TESTING DIFFERENT K VALUES")
for k in [1, 3, 5, 7, 9]:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train_scaled, y_train)
    p = model.predict(X_test_scaled)
    acc = accuracy_score(y_test, p)
    print(f"K = {k} -> Accuracy = {round(acc * 100, 2)} %")

# Visualizations
print("\nGenerating Confusion Matrix Plot...")
fig, ax = plt.subplots(figsize=(7, 5))
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=le_target.classes_)
disp.plot(cmap="Blues", ax=ax)
ax.set_title('K-NN Confusion Matrix (Flight Status)')
plt.tight_layout()
plt.show()
```

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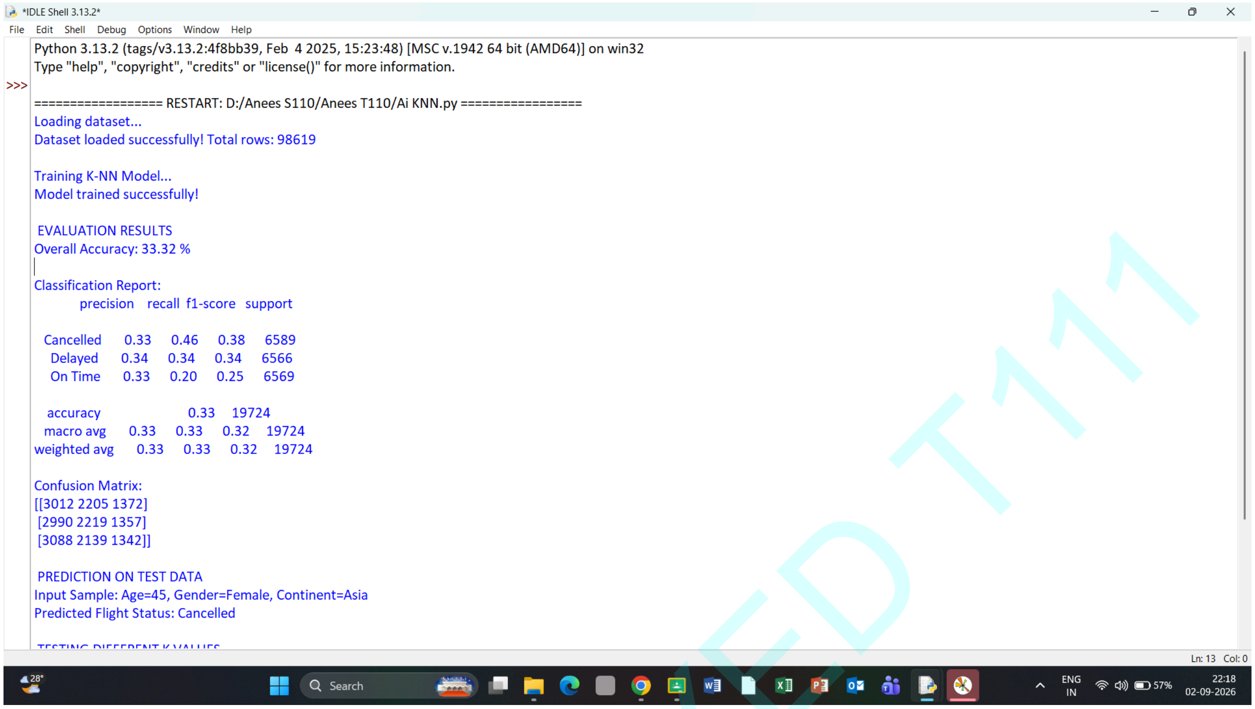
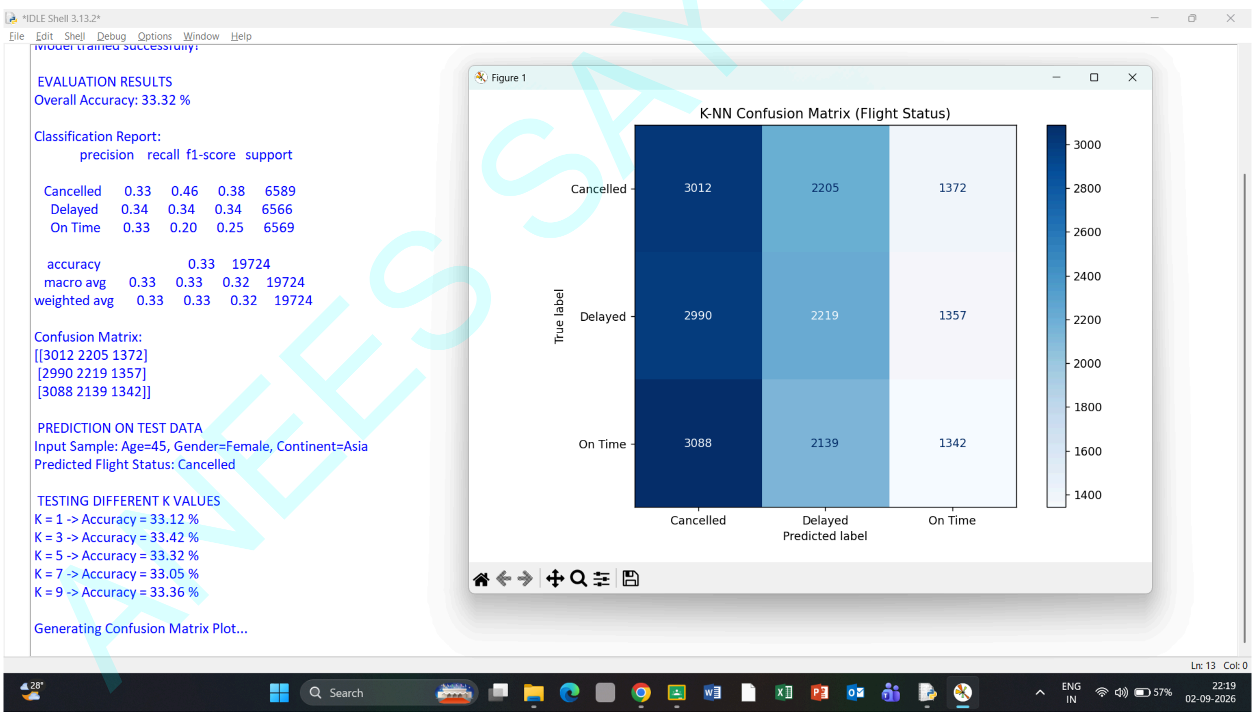
OUTPUT:-



```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================= RESTART: D:/Anees S110/Anees T110/Ai KNN.py ================
Loading dataset...
Dataset loaded successfully! Total rows: 98619

Training K-NN Model...
Model trained successfully!

 EVALUATION RESULTS
Overall Accuracy: 33.32 %

Classification Report:
              precision    recall  f1-score   support

   Cancelled       0.33      0.46      0.38      6589
     Delayed       0.34      0.34      0.34      6566
     On Time       0.33      0.20      0.25      6569

    accuracy                           0.33     19724
   macro avg       0.33      0.33      0.32     19724
weighted avg       0.33      0.33      0.32     19724

Confusion Matrix:
[[3012 2205 1372]
 [2990 2219 1357]
 [3088 2139 1342]]

 PREDICTION ON TEST DATA
Input Sample: Age=45, Gender=Female, Continent=Asia
Predicted Flight Status: Cancelled
```



```
 EVALUATION RESULTS
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Confusion Matrix:
[[3012 2205 1372]
 [2990 2219 1357]
 [3088 2139 1342]]

 PREDICTION ON TEST DATA
Input Sample: Age=45, Gender=Female, Continent=Asia
Predicted Flight Status: Cancelled

 TESTING DIFFERENT K VALUES
K = 1 -> Accuracy = 33.12 %
K = 3 -> Accuracy = 33.42 %
K = 5 -> Accuracy = 33.32 %
K = 7 -> Accuracy = 33.05 %
K = 9 -> Accuracy = 33.36 %

Generating Confusion Matrix Plot...
```


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